SEC P vs NP log 2026-05-19

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 01:40 UTC

SEC P vs NP — daily log 2026-05-19

9 cycles recorded

Decision Tree Depth and Communication Complexity via Lifting

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: DECISION_TREE_DEPTH × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-19 00:11 UTC
- **Entry ID**: `eeec3f948dcf`

Statement

For any Boolean function f on n variables, if the decision tree depth of f is d, then the deterministic communication complexity R_{1/3}(f) is Ω(d / log n).

Rationale

This conjecture explores the relationship between the depth of a decision tree and the communication complexity of the function, leveraging lifting techniques to establish a lower bound. If true, it would imply that functions with deeper decision trees require more communication, which is a key insight in understanding the interplay between these complexity measures.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1302.4207v1] A composition theorem for decision tree complexity - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [0911.3482v5] Complexity of Networks (reprise) - [0101006v1] On Complexity and Emergence

Empirical Test

- exit code: `124`, elapsed: `240.11s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce the parameter grid (e.g., restrict to n ∈ {8,16,24} and sample fewer depths) and/or optimize the deterministic CC computation to fit within the time budget.

Laplacian Apolar Rank Caps Delorme-Poljak Max-CUT SoS-2 Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Curto–Fialkow real moment problem / apolarity & Hankel flat-extension theory (real algebraic geometry). The Hankel rank of the Laplacian moment sequence m_k(G) := tr(L_G^k) is by the Hamburger truncated moment theorem the smallest support size of an atomic measure on R_{≥0} reproducing the moments — equivalently the number of distinct Laplacian eigenvalues (Curto–Fialkow 1996; Lasserre 2009 'Moments, Positive Polynomials and their Applications'; Blekherman–Parrilo–Thomas 2012 'Semidefinite Optimization and Convex Algebraic Geometry'; Reznick 1992). The invariant uses real-eigenvalue clustering and rank-of-Hankel — non-ring (no F_q polynomial extension preserves cluster count of real eigenvalues under polynomial ring lifts), Aaronson–Wigderson algebrization-safe. An arXiv search 'Curto-Fialkow flat extension' AND ('Max-Cut' OR 'Delorme-Poljak' OR 'SoS integrality gap') returns 0 direct hits with <5 adjacent papers (Lasserre hierarchy convergence proofs, Helton–Nie semidefinite representability), none using Hankel-apolar rank of the Laplacian-moment sequence as an SoS-2 Max-CUT tightness invariant. Distinct from blacklisted Lee–Yang zeros of the cut polynomial (complex roots, not real moment-sequence), Fekete-capacity (transfinite diameter of spectrum), free 4-cumulant excess (Voiculescu NC partition statistics), Kashin L^1-flatness (single-eigenvector L$^{1/L}$2 ratio), and Halász L^2 discrepancy (CDF deviation). × Degree-2 Sum-of-Squares / Delorme–Poljak integrality gap ρ(G) := n·λ_max(L_G)/(4·MC(G)) − 1 for the Max-CUT problem on connected k-regular graphs (Goemans–Williamson 1995; Delorme–Poljak 1993; Schoenebeck 2008; Barak–Steurer 2014). The structural form of the Hankel-rank invariant (a count of distinct Laplacian eigenvalues, a property of the GRAPH WIRING — two non-isomorphic graphs with identical Max-CUT value can have very different h(G)) shields it from the Razborov–Rudich natural-proofs barrier; using only real-eigenvalue grouping and integer counting it lies outside the algebrization closure of Aaronson–Wigderson.
- **Recorded**: 2026-05-19 00:24 UTC
- **Entry ID**: `378d9a4dead5`

Statement

For every connected k-regular simple graph G on n ∈ [8,20] vertices with k ∈ {3,4}, let h(G) be the number of distinct nonzero Laplacian eigenvalues of L_G (equivalently the rank of the truncated Hankel matrix H_d(G)_{i,j} = tr(L_G^{i+j+1}) for d ≥ n, the Curto–Fialkow apolar rank of the Laplacian moment sequence). Let MC(G) be the exact Max-CUT value and ρ(G) := n·λ_max(L_G)/(4·MC(G)) − 1 be the Delorme–Poljak SoS-2 integrality gap. Then n · ρ(G) ≤ h(G) · log_2(n+1); a single instance with n·ρ(G) > 2·h(G)·log_2(n+1) refutes the conjecture.

Rationale

Apolarity / Curto–Fialkow theory identifies the Hankel rank of a moment sequence with the minimum number of atoms in a representing measure — a canonical real-algebraic-geometry obstruction. Graphs whose Laplacian spectra collapse onto few atoms (distance-regular, SRG, complete bipartite) tend to admit tight Delorme–Poljak SoS-2 certificates because the symmetry forces the optimal pseudoexpectation onto a low-dimensional invariant subspace. The conjecture promotes the folk 'symmetric spectra ⇒ tight SoS-2' heuristic to a quantitative apolar bound, suggesting Hankel rank as the moment-matrix property P sought by the SOS_DEGREE focus.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [2404.18142v1] Revisiting Majumdar-Ghosh spin chain model and Max-cut problem using variational quantum algorithms - [2411.11721v4] Eigenvalues of the Neumann magnetic Laplacian in the unit disk - [0812.4567v1] On degenerate Hamburger moment problem and extensions of positive semidefinite Hankel block matrices - [1508.00617v2] Hankel determinants of random moment sequences

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 123, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 96, in run_trial
    eigenvalues = laplacian_eigenvalues(graph)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 58, in laplacian_eig
    eigenvalues = gaussian_elimination(L)
                  ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_06da5d96.py", line 29, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in the custom gaussian_elimination routine before any trials completed, so no r-values were produced and neither the support nor falsification condition can be evaluated. | next: Replace the hand-rolled Gaussian elimination with numpy.linalg.eigvalsh (or use pivoting) to robustly compute Laplacian eigenvalues, then rerun the 240-seed sweep.

Cross-Read Kronecker Sum Rank Lower-Bounds Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: GCT-style plethystic Kronecker-product orbit-tangent obstructions via the bit-flip operators D_p := T_p^1 - T_p^0 in Z^{w x w} (infinitesimal GL(w)-tangent directions to the BP's orbit-closure in the Mulmuley-Sohoni 2001 / Burgisser-Ikenmeyer 2013 sense), aggregated as the cross-read Kronecker sum K(P) := sum_v D_{p_1(v)} (x) D_{p_2(v)} in Z$^{\{w}$2 x w^2}. The invariant is STRUCTURAL on the BP wiring (two read-twice BPs computing the same function can have very different K), shielding from Razborov-Rudich natural-proofs; rank over Z has no canonical F_q polynomial lift preserving the integer rank value across the Kronecker sum (the rank of a Kronecker SUM is not preserved by polynomial ring lifts, unlike the rank of a single Kronecker product), so the bridge is Aaronson-Wigderson algebrization-safe. An arXiv search 'Kronecker sum bit-flip' AND ('read-twice branching program' OR 'Borodin Razborov Smolensky' OR 'GCT plethysm') returns 0 direct hits with <5 adjacent papers (all on Kronecker-coefficient plethysm computations or layer-product BP lower bounds, never on the cross-read SUM of bit-flip Kronecker products). Distinct from blacklisted Generated-Algebra-Dimension (Wedderburn span via Gauss-Jordan closure), Schatten Stable-Rank of Layer-Difference Stack (Schatten 2/inf ratio on a vertically stacked layer-difference matrix, NOT a Kronecker sum), Cycle-Type Mixing (Plancherel S_5 distance), and Coxeter-Inversion Spread (S_5 reflection-length word statistic) - none uses the cross-read Kronecker SUM of bit-flip operators. × Read-twice oblivious branching program size s(P) for IP_2(x,y) = XOR_k (x_k AND y_k) on 2n inputs under the adversarial variable order x_1,…,x_n,y_1,…,y_n,x_1,…,x_n,y_1,…,y_n (Borodin-Razborov-Smolensky 1993 regime), instantiating the focus's 'distinguishing tensor' formalism with rho(P)=log_2 rank K(P) = O(log s(P)) but rho(any RT-BP for IP_2) = Omega(n).

- **Recorded**: 2026-05-19 00:32 UTC
- **Entry ID**: `fd57783d2ed4`

Statement

Let P be a read-twice oblivious BP with width w and 0/1-valued layer transition matrices T_p^b in {0,1}^{w x w} for layers p in [4n] and bits b in {0,1}, with each variable v read at exactly two positions p_1(v) < p_2(v). Define the bit-flip operator D_p := T_p^1 - T_p^0 in Z^{w x w} and the cross-read Kronecker sum K(P) := sum over all 2n variables v of D_{p_1(v)} tensor D_{p_2(v)} in Z$^{\{w}$2 x w^2}, and let rho(P) := rank_Q(K(P)). Conjecture: (a) rho(P) <= w(P)^2 <= s(P)^2 for every read-twice oblivious BP P (trivial dimension cap), and (b) for every read-twice oblivious BP P computing IP_2 under the adversarial order above, rho(P) >= 2^n / 4. Equivalently, any read-twice oblivious BP for IP_2 has size at least 2^{n/2}/2; a single (P,IP_2) pair with rho(P) < 2^n/4 falsifies (b).

Rationale

In Mulmuley-Sohoni GCT the orbit-closure of a polynomial is probed by GL(w)-equivariant tangent directions; the bit-flip operators D_p are exactly the layer-wise infinitesimal generators of the BP's representation variety, and their cross-read Kronecker product encodes the plethystic coupling that READ-ONCE BPs cannot create. For IP_2 under adversarial order, each variable v forces the two reads at p_1(v),p_2(v) to jointly implement an AND-XOR coupling: when bits are flipped the resulting D_{p_1(v)} tensor D_{p_2(v)} term must contribute a fresh rank-1 block to K(P), and the n distinct variables produce 2n rank-1 blocks whose supports cannot algebraically cancel without enlarging the width. The fooling-set / matrix-rank machinery is non-ring and structural on the BP wiring (not the truth table), shielding the bridge from both Razborov-Rudich natural proofs and Aaronson-Wigderson algebrization.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1601.02745v1] Basic Reasoning with Tensor Product Representations - [1511.07136v1] Identity Testing and Lower Bounds for Read-k Oblivious Algebraic Branching Programs - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [2107.09834v4] Communication lower bounds for nested bilinear algorithms via rank expansion of Kronecker products

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 91, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 61, in run_trial
    rho_ip2 = compute_rho(n)
              ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 28, in compute_rho
    P_IP2 = build_read_twice_bp(n)
            ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 23, in build_read_tw
    D_p.append([[T_p_1[i][j] - T_p_0[i][j] for j in range(w)] for i in range(w)])
                 ~~~~~~~~^^^
TypeError: 'int' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError in build_read_twice_bp before producing any rho data, so neither the support nor falsification condition could be evaluated. | next: Fix the BP construction bug (ensure T_p_0 and T_p_1 are 2D w×w matrices, not ints) and rerun the pre-registered protocol for n in {2,3,4,5}.

SoS Cone-Gram Stable Rank Caps AC^0 Depth-d Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares hierarchy / Lasserre stable-rank geometry of structural Gram matrices on cone-indicator features (Lasserre 2001; Parrilo 2003; Barak–Steurer 2014; Schur 1923 Hadamard-product PSD theory; Rudelson–Vershynin 2007 stable rank). The invariant ψ(C) := Σλ_i(K(C)) / λ_max(K(C)) — the Schatten-1/Schatten-∞ stable rank of the gate-cone-indicator Gram matrix K(C)[g,h]=|cone(g)∩cone(h)|/n — is the SoS-degree-2 second-moment / pseudoexpectation functional of the structural feature ensemble {1_{cone(g)}/√n}_{g∈V(C)} viewed as a Lasserre relaxation of the gate-incidence polytope. It is a non-ring, real-spectral Schatten-ratio (no F_q polynomial extension preserves Σλ/λ_max under polynomial ring lifts), Aaronson–Wigderson algebrization-safe. STRUCTURAL on the circuit DAG (two AC^0 circuits with the same Boolean function can have very different cone Gram spectra), so it sidesteps the Razborov–Rudich barrier that killed the prior 'Lasserre Moment-Rank Gap of Accept Set' attempt (which operated on truth-table accept sets). Sym(n)-equivariant under input permutations (K(C) transforms by simultaneous permutation, preserving spectrum), fulfilling the focus's natural-group-action requirement. An arXiv search 'Lasserre stable rank' AND ('AC^0' OR 'PARITY lower bound' OR 'switching lemma') returns 0 direct hits with <5 adjacent papers (all on SoS-Max-Cut / SoS-CSP integrality gaps, never on gate-cone Gram spectra of AC^0 circuits). × AC^0 circuit size s(C) and depth d for the explicit n-bit PARITY function, in the Håstad 1986 switching-lemma / Razborov–Smolensky / Beame 1994 regime, accessed through the depth-d $2^{\Theta(n}${1/(d−1)})} lower bound (open: tight constants and a non-restriction-based proof). In the Cook–Reckhow–Krajíček bounded-arithmetic framework, an AC^0 PARITY size lower bound is the canonical V^0 □ V^0(⊕) separation stepping stone.
- **Recorded**: 2026-05-19 00:39 UTC
- **Entry ID**: `63e0e3f43696`

Statement

For every AC^0 circuit C of depth d ≥ 2 on n input variables that computes PARITY_n exactly, the gate-cone-indicator Gram matrix K(C)∈R^{|V|×|V|} with K[g,h]:=|cone(g)∩cone(h)|/n satisfies ψ(C):=(Σ_i λ_i(K(C)))/λ_max(K(C)) ≤ C_0·($\log_2$ s(C))^{d−1}, equivalently $\log_2$ s(C) ≥ $\psi(C)^{1/(d-1)}/C_0${1/(d−1)}, for an absolute constant C_0 ≤ 4. The bound is non-vacuous because PARITY requires every input in cone(output), forcing ψ(C) to scale at least as n^{(d−2)/(d−1)} on any depth-d AC^0 realization, matching Håstad's lower-bound shape without invoking random restrictions. A single AC^0 circuit C computing PARITY with ψ(C)^{1/(d−1)} > 4·$\log_2$ s(C) falsifies the conjecture.

Rationale

The cone Gram matrix K(C) is the natural SoS-degree-2 Lasserre moment-matrix of the structural feature ensemble {1_{cone(g)}}_g, and its stable rank ψ(C) is a Schur–Horn-type combinatorial second-moment measuring how many 'effective independent input directions' the wiring of C must explore — exactly the resource that Håstad's switching lemma shows PARITY forces large via random restrictions, but here computed deterministically from the DAG alone. Because cone(g) depends only on the wiring (not the gate functions or truth table), two AC^0 circuits with identical Boolean function can have very different ψ(C), giving a Razborov–Rudich-safe structural functional within the SoS hierarchy. The bound's exponent (d−1) mirrors Håstad's tight $2^{\Theta(n}${1/(d−1)})} shape, so if it holds even loosely it yields a restriction-free route to AC^0 PARITY lower bounds and a candidate ψ for the focus's polynomial-time natural-invariant target.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 151, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 108, in run_trial
    cones = [reverse_bfs(g, i) for i in range(n)]
             ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 28, in reverse_bfs
    if not visited[neighbor]:
           ~~~~~~~^^^^^^^^^^
TypeError: list indices must be integers or slices, not tuple
```

Judge reasoning

The test crashed with a TypeError in reverse_bfs (list indexed by a tuple) before producing any r(C) values, so neither the support nor falsification condition could be evaluated. | next: Fix the cone-construction bug in reverse_bfs (ensure neighbor is an integer node id, not a (node, edge) tuple) and rerun the 540-instance sweep.

Talagrand L1 Influence Spread of Clause-Falsification Polynomial Lower-Bounds Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Talagrand L1-influence-vector / KKL-Friedgut Boolean Fourier analysis — the Talagrand 1994 functional T(p) := Σ_i √Inf_i(p), with Inf_i(p) = Σ_{S ∋ i} p̂(S)², applied as a SYNTACTIC Fourier-analytic invariant of the clause-falsification pseudo-Boolean polynomial p_F(x) := #{C ∈ F : x falsifies C} of a CNF (a property of the clause LIST — multiplicities and polarities — not of the truth table; two unsat 3-CNFs with identical satisfiability function ≡ FALSE can have very different T(F)). Distinct from blacklisted Mansour L1 mass Σ_S |p̂(S)| (a coefficient-indexed absolute-value sum), Bonami 4/2 ratio ||p||_4/||p||*2 (a hypercontractive norm ratio), Bourgain NSM Σ_S (1−ρ^{|S|}) p̂(S)² (a level-weighted L² mass), hypercontractive term-pair noise log-ratio (a Rényi-2-to-4 kernel statistic), and cross-side Fourier symmetric-difference mass (a truth-table communication-matrix Fourier object that hit NATURAL_PROOFS) — Talagrand L1 is the unique coordinate-indexed Σ_i √(Σ*{S∋i} p̂(S)²) functional with an OUTER √ over per-variable L² influence buckets. The functional uses √ on integer-valued partial sums and a per-coordinate sum that does NOT extend to a polynomial F_q lift (square-root of an L² influence has no polynomial-ring surrogate), Aaronson–Wigderson algebrization-safe; structural shielding from Razborov–Rudich since T(F) is not a poly-time-computable property of the truth table of F (which is ≡ FALSE). ArXiv 'Talagrand L1 influence sum' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (KKL/Friedgut juntas, never on syntactic clause-falsification polynomials). × Tree-like Resolution / DPLL refutation size t*(F) for unsatisfiable 3-CNFs F in the Ben-Sasson–Wigderson 2001 / Beame–Karp–Pitassi–Saks 2002 / Krajíček 1995 bounded-arithmetic regime where ¬F is*

Σ^b_0 and log_2 t(F) lower-bounds V^0-witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák–Buss feasible interpolation.
- **Recorded**: 2026-05-19 00:46 UTC
- **Entry ID**: `9e12285a05e1`

Statement

For every unsatisfiable 3-CNF F on n ≤ 40 variables with m clauses, let p_F : {0,1}^n → Z_{≥0} count falsified clauses, expand p_F in the Walsh basis (only subsets S with |S| ≤ 3 are non-zero), and define T(F) := Σ_{i=1}^n √(Σ_{S ∋ i, 1 ≤ |S| ≤ 3} p̂_F(S)²). Then log_2(1 + t*(F)) ≥ 0.10 · T(F) / √m, where t*(F) is the DPLL refutation size (backtrack count) under the unit-propagation + first-unassigned variable order. A single F violating this inequality (i.e. log_2(1+t*(F)) < 0.10 · T(F)/√m) refutes the conjecture.

Rationale

Clause-falsification influence Inf_i(p_F) measures how strongly variable i 'pivots' the count of broken clauses, and Talagrand's L^1-spread functional Σ_i √Inf_i interpolates between the variance Σ_i Inf_i (= I(p_F), the total clause-sensitivity) and the maximum influence KKL bound, capturing per-variable contention without collapsing to a coefficient sum (Mansour) or a hypercontractive norm ratio (Bonami). Resolution / DPLL must spend ≥ Ω(width) steps to refute, and Ben-Sasson–Wigderson's width-size tradeoff together with Talagrand's variance-spread inequality suggest the clause-list-side T(F) functional should track refutation width, hence log t*. The bridge is structural (not a truth-table property), uses non-polynomial operations (√, |·|), and lies outside both the Razborov–Rudich constructivity regime and the Aaronson–Wigderson algebrization closure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 175, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 132, in run_trial
    if not is_unsatisfiable(cnf):
           ^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in is_unsatisfia
   n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.repla
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in <genexpr>
   n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.repla
                                                            ^^^^^^^^^^^^^^^^

ValueError: invalid literal for int() with base 10: ''
```

Judge reasoning

The test crashed with a ValueError during CNF parsing (is_unsatisfiable) before any instances were evaluated, so no violation count or Pearson correlation was produced. The pre-registered support condition cannot be assessed. | next: Fix the CNF literal parser in is_unsatisfiable (handle the negation prefix and empty tokens correctly) and rerun the pre-registered sweep over (n,α) with 30 seeds.

Border Rank Lower Bound for Communication Complexity of Disjointness

- **Verdict**: INCONCLUSIVE
- **Bridge**: ALGEBRAIC_GEOMETRY (border rank of tensors) × COMMUNICATION_COMPLEXITY
- **Recorded**: 2026-05-19 01:09 UTC
- **Entry ID**: d76acad80bde

Statement

The randomized communication complexity of the DISJ_n matrix is Ω(log(border_rank(M))), where border_rank(M) is the border rank of the communication matrix over the real numbers.

Rationale

The border rank of a tensor captures its asymptotic complexity, and higher border ranks imply greater information-theoretic requirements for communication. This conjecture links algebraic geometry's tensor rank framework to communication complexity, leveraging the known Ω(n) lower bound for DISJ_n as a baseline.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1608.07486] A $2n^2 - log(n) - 1$ lower bound for the border rank of matrix multiplication - [RNX025] A $2\mathbf{n}^2 - \log_2(\mathbf{n}) - 1$ lower bound for the border rank of matrix multiplication

Empirical Test

- exit code: 0, elapsed: 0.20s

```
_value': 24, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 24, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 15, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 12, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 4, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 22, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 20, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 6, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 35, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 13, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 38, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 16, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 14, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=19.5 std=10.465021102065045 support_fraction=1.0
```

Judge reasoning

The test output only reports a single 'communication_complexity' integer per trial with no border_rank values shown, so the pre-registered criterion (comparing R(DISJ_n) to log2(border_rank_mean)) cannot be verified; even if computed, the bound log(border_rank) ≤ n is trivially satisfied by R(DISJ_n)=Θ(n), making the 'SUPPORTED' result vacuous. | next: Re-run with explicit border_rank values logged per n, and compare against a nontrivial benchmark function where border rank is provably super-pol

Communication Complexity Lower Bound for Disjointness Implies Monotone Circuit Size Lower Bound

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: COMMUNICATION_COMPLEXITY × MONOTONE_CIRCUIT_LOWER_BOUNDS
- **Recorded**: 2026-05-19 01:23 UTC
- **Entry ID**: `bf58b88b218f`

Statement

For the disjointness function on n-bit inputs, if the deterministic communication complexity is Ω(n), then every monotone circuit computing it requires size Ω(2^{n/2})

Rationale

The Karchmer-Wigderson game links communication protocols to circuit depth, but monotone circuits require stronger lower bounds. Disjointness has known Ω(n) communication complexity, and its monotone circuit size lower bound remains open. A direct connection would unify these complexity measures via combinatorial geometry of protocols.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:2506.12420] Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication - [3406325.3451069] Lower bounds for monotone arithmetic circuits via communication complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 89, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 67, in run_trial
    D = compute_communication_complexity()
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 46, in compute_commu
    protocol_tree_search(inputs, 0)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
   [Previous line repeated 1 more time]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 44, in protocol_tree
    protocol_tree_search(right_partition, depth + 1)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree
    protocol_tree_search(left_partition, depth + 1)
  [Previous line repeated 989 more times]
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

The test crashed with a RecursionError in protocol_tree_search before producing any data, so neither the communication complexity D nor the monotone circuit size S was measured. With no metrics computed and the critic challenging, the pre-registered support condition cannot be evaluated. | next: Rewrite protocol_tree_search iteratively (or cap depth at 2n and prune by rectangle monochromaticity) so D(DISJ_4) and S(DISJ_4) can actually be computed before re-running the 10-seed trial.

Cubical First-Betti Linearly Lower-Bounds DNF-MCSP

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Cubical combinatorial topology — the cubical subcube complex K_f := {σ⊆{0,1}^n : σ is a subcube and σ⊆f^{-1}(1)} with its boundary maps over F_2, β_1(K_f) computed as nullity(∂_1)−rank(∂2) *(Kaczynski–Mischaikow–Mrozek 2004 'Computational Homology'; Edelsbrunner–Letscher–Zomorodian 2002; Carlsson 2009 'Topology and data'). A non-ring, F_2-rank-of-cubical-boundary functional: ranks of integer cellular boundaries do NOT extend to a polynomial F_q lift of the Boolean ring (the cubical boundary involves signed face inclusion not preserved by polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. The invariant is RELATIONAL (a categorical functor (Bool^n → Top → Vect*{F_2}) applied to the subcube cover poset), not a single truth-table number — sidestepping trap (j). An arXiv search 'cubical homology' AND ('MCSP' OR 'minimum DNF' OR 'meta-complexity') returns 0 direct hits with <5 adjacent papers (all on TDA of biological data or persistent-homology classifiers, never on DNF-MCSP). Distinct from blacklisted Mahler-measure (single complex-analytic truth-table number), FCA antichain width (Galois-lattice width on assignments×literals), treewidth of PI compatibility graph (graph-minor invariant), and independence-complex Euler χ̃ (Stanley–Reisner alternating sum). × DNF-MCSP / minimum DNF size σ_DNF(f) for f:{0,1}^n→{0,1} in the Quine–McCluskey / Masek 1979 / Umans 1999 / Allender–Hellerstein–McCabe–Pitassi–Saks / Hirahara–Oliveira meta-complexity regime, where σ_DNF is the optimum of a prime-implicant set-cover and the canonical NP-hard meta-complexity object whose hardness fuels Hirahara's worst-case-to-average-case program. Sparsity escape from Razborov–Rudich: 'σ_DNF(f) ≤ s' is rare among random f (only ~2^{O(s log n)} functions admit it, of $2^{\{2}}$n} total), so any lower bound σ_DNF ≥ φ(f) inherits sparsity from the cap side and does not violate Razborov–Rudich largeness.
- **Recorded**: 2026-05-19 01:33 UTC
- **Entry ID**: `9020e29f840c`

Statement

Let K_f = {σ ⊆ {0,1}^n : σ is a subcube and σ ⊆ f^{-1}(1)} be the cubical subcube complex of f^{-1}(1), and β_1 := β_1(K_f, F_2) its first F_2-Betti number. Then for every Boolean f:{0,1}^n→{0,1} satisfying β_1(K_f) ≥ 1, the minimum DNF size obeys σ_DNF(f) ≥ 3·β_1(K_f). One counterexample (a single f with σ_DNF(f) < 3·β_1(K_f) and β_1(K_f) ≥ 1) refutes the conjecture.

Rationale

By the nerve theorem applied to any DNF cover F = T_1∨…∨T_s of f (each T_i a subcube ⊆ f^{-1}(1)), the nerve N_F is homotopy equivalent to K_f, giving β_1(N_F)=β_1. Trivially N_F has s vertices and any 1-cycle uses ≥3 of them, yielding only σ_DNF ≥ √(2β_1); the conjecture asserts the much stronger LINEAR bound — each independent topological hole costs 3 fresh terms in any optimal cover because subcubes encircling a hole are 'rigid' (cannot be shared across distinct holes without collapsing the cycle). This is a clean nerve-rigidity strengthening pioneered by no prior MCSP work.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 105, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 84, in run_trial
    beta_1, sigma_DNF = dnf_size(f, n)
                        ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 25, in dnf_size
    beta_1 = compute_beta_1(K_f)
             ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 30, in compute_beta_
    n = len(K_f[0])
            ~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in compute_beta_1 (empty K_f list) before any instances were evaluated, so no data was produced to assess the pre-registered support or falsification condition. | next: Fix compute_beta_1 to handle the empty subcube complex case (β_1 = 0 when K_f is empty) and rerun the 270-triple sweep.

Fourier Spectral Entropy of Clause-Falsification Polynomial Bounds Tree-Resolution

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier-analytic information theory — Shannon entropy of the normalized Walsh–Hadamard spectrum of a multilinear pseudo-Boolean polynomial (Friedgut–Kalai 1996 FEI conjecture; O'Donnell 2014 'Analysis of Boolean Functions' Ch. 9; Wan–Wang–Yu 2014 on FEI for low-degree functions; Chakraborty–Kulkarni–Lokam–Saurabh 2016 'Upper bounds on Fourier entropy'). The functional H_F(F) := $-\Sigma$_S q_F(S)$\cdot\log_2$ q_F(S), where q_F(S) := p̂_F(S)^2/||p_F||^2, is the Shannon entropy of the SPECTRAL probability distribution viewed as a measure on the Boolean character lattice 2[1]. It is distinct from Mansour L^1 Σ|p̂(S)| (linear in |p̂|, no log), Bourgain NSM Σ_S(1−ρ^|S|)p̂(S)^2 (level-indexed L^2 weight, not entropy), Bonami 4/2 ratio $||p||_4/||p||_2$ (norm ratio), Talagrand Σ_i √Inf_i (per-coordinate L^2-root), term-pair noise log-ratio (Rényi-2-to-4 hypercontractive kernel), and cross-side symdiff mass (truth-table CC-matrix Fourier). It uses log of a normalized ratio of squares, so it admits no polynomial F_q extension (Shannon entropy of a probability distribution is not preserved under polynomial-ring lifts of the Boolean ring), making the bridge Aaronson–Wigderson algebrization-safe. Razborov–Rudich-safe because H_F(F) is a property of the CLAUSE LIST (multiplicities,

[1] n

polarities), so two unsat 3-CNFs F_1,F_2 with identical satisfiability function (both ≡ FALSE) can have very different H_F. An arXiv search 'Fourier entropy' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation' OR 'proof complexity') returns 0 direct hits and <5 adjacent papers (all on FEI for ±1 Boolean f or random-DNF entropy, never on the clause-falsification polynomial of an unsat CNF). × Tree-like Resolution / DPLL refutation size t*(F) for unsatisfiable 3-CNFs F in the Chvátal–Szemerédi 1988 / Ben-Sasson–Wigderson 2001 / Beame–Karp–Pitassi–Saks 2002 / Krajíček 1995 bounded-arithmetic regime, where ¬F is Σ^b_0 and $\log_2$ t*(F) lower-bounds V^0-witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák–Buss feasible interpolation.
- **Recorded**: 2026-05-19 01:40 UTC
- **Entry ID**: `363fda00c17f`

Statement

Let F be an unsatisfiable 3-CNF on n variables with m clauses. Define p_F : {−1,+1}^n → ℝ by p_F(x) := Σ_{C∈F} 𝟙[x falsifies C]; its Walsh–Hadamard spectrum p̂_F(S) is supported on |S|≤3 and admits the closed form p̂_F(∅)=m/8, p̂*F({i})=−Σ*{C∋i}ε_i^C/8, p̂*F({i,j})=Σ*{C⊇{i,j}}ε_i^C ε_j^C/8, p̂*F({i,j,k})=−Σ*{C={i,j,k}}ε_i^C ε_j^C ε_k^C/8 where ε_i^C∈{±1} is the polarity of variable i in clause C. Let q_F(S):=p̂*F(S)²/||p_F||² and H_F(F):=−Σ*{S:|S|≤3} q_F(S)·$\log_2$ q_F(S). Conjecture: there exists a universal constant c>0 such that for every unsat 3-CNF F at clause-to-variable ratio α∈[4.2, 5.0], $\log_2$ t*(F) ≥ c · n · H_F(F) / $\log_2(1+n+n^2+n^3)$. One counterexample (an unsat 3-CNF whose ratio $\log_2$ t*(F) · $\log_2(1+n+n^2+n^3)$ / (n·H_F(F)) is arbitrarily small) refutes it.

Rationale

The spectral entropy H_F measures how DELOCALIZED the Walsh–Hadamard energy of the clause-falsification polynomial is across degree-≤3 character modes; intuitively, a CNF whose Fourier mass concentrates on a few low-degree characters admits a short tree-Resolution refutation (a few decisions kill the dominant modes), while a CNF whose Fourier mass spreads broadly forces DPLL to branch in many directions. The Friedgut–Kalai FEI heuristic — entropy is bounded by total influence — suggests an inverse linear relationship between spectral concentration and combinatorial complexity, and here we are normalizing entropy by the log of the number of degree-≤3 characters to get an n-scaled functional. The bridge is genuinely under-explored: spectral entropy of the syntactic clause polynomial has never been used as a DPLL lower-bound certificate.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 139, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 111, in run_trial
    spectrum = walsh_hadamard_transform(cnf)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 59, in walsh_hadamar
    spectrum = [0] * (2 ** n_vars)
               ~~~~^~~~~~~~~~~~~~~
OverflowError: cannot fit 'int' into an index-sized integer
```

Judge reasoning

The test crashed with an OverflowError when attempting to allocate a 2^n-sized array for the Walsh-Hadamard transform, so no R(F) values or Pearson correlations were produced. Without data, neither the support nor the falsification condition can be evaluated. | next: Rewrite walsh_hadamard_transform to exploit the closed-form spectrum (support only on $|S|\leq 3$, $O(n^3)$ coefficients) rather than allocating a 2^n array, then rerun the pre-registered protocol.